
Beyond IID: Controlled Studies of Visual Cues, Domain Alignment, and Open-Set Rejection

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Code and artifacts: github.com/mujeeb-asad/ATML-PA1

Abstract

Four settings in which the closed-set IID assumption fails are examined under one shared protocol: controlled image interventions on frozen backbones, unsupervised domain adaptation and domain generalization on PACS, and open-set recognition on CIFAR. The interventions show that colour and position are nearly costless for all three backbones while destroying global spatial arrangement is expensive, and that the models divide by pretraining rather than by architecture. On PACS, no adaptation method improves on the source-only baseline at its specified setting, and the two adversarial methods leave domain separability unchanged, so domain invariance does not account for their behaviour; a class probe recovers most of the class structure from the same frozen features, placing the failure in the classifier rather than the representation. With the target domain withheld entirely, both domain-generalization methods improve on that baseline, so access to unlabelled target data was not the deciding factor. In open-set recognition a classifier at 94.9% closed-set accuracy still accepts roughly two thirds of semantically near unknowns at a calibrated operating point, and stronger augmentation improves rejection without improving accuracy. Across the four tasks, measurements taken at one level of a model repeatedly fail to license claims at another. Code, configurations and machine-readable results: <https://github.com/mujeeb-asad/ATML-PA1>.

1 Introduction

Supervised learning normally assumes that training and test data share a distribution and a label space. Deployment violates both. An object may appear with unfamiliar texture or in an unfamiliar rendering style; a test image may come from a domain never seen during training; an input may belong to no training class at all. A model can be accurate in-distribution and still fail under each of these, because accuracy alone does not reveal *which* evidence produced it.

This report examines four such settings. Task 1 applies controlled image interventions to three frozen backbones and asks which visual cues drive their predictions, and whether changes in prediction track changes in the underlying feature. Task 2 gives a model unlabelled examples from a target domain and asks whether aligning source and target representations improves recognition there. Task 3 removes the target domain entirely and compares source-domain alignment against parameter-space stability. Task 4 asks when a classifier should decline to predict at all.

Section 2 gives the shared protocol and the deviations from it; Section 7 relates the four sets of results; Section 8 states what a single seed permits.

2 Experimental Setup

All random choices use seed 6304 and every experiment ran once, so no result carries an error bar. Checkpoints are the best epoch under a fixed rule—mean source-validation macro-F1 (Tasks 2–3), CIFAR-10 validation accuracy (Task 4).

Datasets and models. *Task 1:* STL-10, stratified 80/20 training split, class-balanced 500-image test subset. Three frozen backbones—ResNet-50 (2048-d pooled), ViT-B/16 (768-d class token), OpenCLIP ViT-B/32 (512-d normalised)—each read by a linear head trained on cached features with AdamW (10^{-3} , ≤ 50 epochs, patience 5); zero-shot CLIP uses the prompt *a photo of a {class}*. Interventions are built on one 224×224 uint8 canvas with per-model normalisation applied afterwards, so all four readouts see byte-identical images. *Tasks 2–3:* PACS, Photo/Art/Cartoon as sources and Sketch as target, sharing one protocol—identical per-domain 80/20 splits loaded from a saved index file, ResNet-18 with a seven-class head, $256 \rightarrow 224$ random crop and flip, AdamW (10^{-4} , ≤ 30 epochs, patience 5), domain-balanced batches of eight per source domain, plus 24 unlabelled target images per step in Task 2. BatchNorm running statistics stay frozen at pretrained values with γ, β trainable, so no method gains implicit adaptation through them. Task 3 loads Task 2’s source-only checkpoint as ERM; evaluated identically the two agree to 0.0000 points. *Task 4:* CIFAR-10 under a stratified 90/10 split gives ten known classes, and sixteen fixed CIFAR-100 test classes give eight near and eight far unknowns of 800 images each; all models use the same CIFAR ResNet-18 with SGD (0.1, cosine, batch 128, 100 epochs), and one cache of logits and features per model serves all post-hoc scores.

Leakage locks. Forbidden access fails rather than silently succeeding: Task 2’s target dataset is built with `labels=None`, Task 3 nulls every Sketch image until the final cell, and Task 4 downloads only the CIFAR-100 test split. Each notebook marks a locked-evaluation boundary with training, selection and source-side diagnostics above it, and qualitative examples are drawn by seeded shuffle. Runs executed on one NVIDIA RTX PRO 6000 under torch 2.11.0+cu128 in fp32; `cuda.deterministic` is not set, so runs are not bitwise reproducible.

Deviations. Ten decisions depart from the specification. Three are stabilisation measures under the course’s authorisation to use clipping or normalisation with documentation. **(1)** Gradient clipping at norm 1.0 in every Task 2–3 run, prompted by a source-side instability; the threshold is arbitrary and the fraction of steps it binds on is logged. **(2)** The MMD estimator excludes self-pairs: the specification fixes the kernel but not the estimator, and on samples from a *single* distribution the biased form reads 0.473 at eight per side and 0.157 at twenty-four—in Task 3 larger than the discrepancy measured—so small negative values are expected. **(3)** The discriminator input is L2-normalised for DANN and CDAN, since binary cross-entropy is unbounded above and a backbone can otherwise reduce the domain loss by inflating feature magnitude, which frozen BatchNorm cannot absorb; on a fixed batch the raw loss rises from 0.70 to 145.8 under a $1000\times$ scaling while the normalised loss is invariant. The classifier head still reads the unnormalised feature and DAN is unchanged; Section 4 returns to the consequence.

The remainder: **(4)** a `StandardScaler` precedes the separability probe, lbfgs converging poorly on raw post-ReLU features; **(5)** CIFAR is read from pinned HuggingFace mirrors rather than torchvision for download speed, so split membership differs and Task 4 is not bit-comparable with a torchvision run; **(6)** PROSER’s detection score fixes the dummy-column bias at zero, the operating point being set by a different rule here; **(7)** an additional DANN run at domain-loss weight 0.1 was trained with the instructor’s approval, the main comparison retaining unit weight; **(8)** more than one bounded study was therefore run, the λ_{MMD} sweep being the designated one and the domain-weight arm a diagnostic; **(9)** a label-shift bound and a class probe were added after the target lock; **(10)** a hue-rotation row appears in Task 1’s stability table beyond the four required transformations. Values fixed by neither the specification nor a principle: AdaIN $\alpha = 0.8$, the 0.35 edge-correlation floor, 24 conflicts per ordered class pair, 50 PCA components before t-SNE, 16 px as the stability displacement, and the clip norm.

Table 1: Task 1 clean, grayscale, hue-rotated and patch-shuffled performance on the same 500 images. Δ Top-1 is against each model’s own clean baseline; consistency is the share of predictions unchanged. Confidence is comparable within a model, not across.

Condition	Model	Top-1 (%)	Δ Top-1	Macro-F1	Consistency (%)
clean	resnet50 head	97.60	0.00	97.61	–
clean	vit_b_16 head	97.40	0.00	97.39	–
clean	clip head	97.60	0.00	97.59	–
clean	clip zero shot	96.80	0.00	96.75	–
grayscale	resnet50 head	95.60	-2.00	95.59	96.40
grayscale	vit_b_16 head	95.60	-1.80	95.59	95.20
grayscale	clip head	95.60	-2.00	95.57	95.60
grayscale	clip zero shot	94.20	-2.60	94.11	95.20
hue rotated	resnet50 head	93.20	-4.40	93.25	94.60
hue rotated	vit_b_16 head	94.20	-3.20	94.18	94.60
hue rotated	clip head	95.00	-2.60	94.94	95.60
hue rotated	clip zero shot	93.40	-3.40	93.31	94.20
patch shuffled	resnet50 head	89.80	-7.80	89.81	92.00
patch shuffled	vit_b_16 head	89.00	-8.40	88.89	90.00
patch shuffled	clip head	79.00	-18.60	78.94	79.00
patch shuffled	clip zero shot	78.80	-18.00	78.86	80.20

Table 2: Cue-conflict decisions over 204 conflicts. Shape bias divides by decided images only; coverage gives the share decided. Raw counts are shown because the denominators differ sharply.

Model	Shape	Texture	Other	Shape bias (%)	Coverage (%)
resnet50 head	65	31	108	67.71	47.06
vit_b_16 head	109	22	73	83.21	64.22
clip head	121	5	78	96.03	61.76
clip zero shot	119	6	79	95.20	61.27

3 Task 1: Inductive Biases and Feature Representations

Three frozen backbones—ResNet-50, ViT-B/16 and CLIP ViT-B/32—are read by linear heads, with zero-shot CLIP as a fourth readout of the CLIP embedding. All four sit within 0.8 points on clean images, so the quantity of interest is the fall under each intervention. The ceiling sets the scale: at 97.6% twelve of 500 images are wrong, so two points is ten images.

Colour. Removing colour costs about two points for every model, with a 0.8-point spread across the four—four images, noise. Rotating hue by half a turn costs *more* than removing it for all four: absent colour is missing evidence, rotated colour is evidence pointing the wrong way. Only ResNet’s gap exceeds noise, 4.4 against 2.0 points.

3.1 Shape versus texture

Cue-conflict images are generated by AdaIN style transfer [Huang and Belongie, 2017]; predictions split into content class, style class, or neither, and only the first two are scored, with shape bias and coverage as defined by Geirhos et al. [2019]. Every model prefers shape, ordered ResNet < ViT < CLIP. ResNet reaches 67.7% and ViT 83.2%; their Wilson intervals overlap by 0.3 points, but interval overlap is over-conservative for two proportions, and a two-proportion test on 65/96 against 109/131 gives $z = 2.73$, $p = 0.006$.

Coverage bounds the conclusion. ResNet decides only 47.1% of conflicts, so its figure describes 96 of 204 images and the 108 undecided ones leave its shape share over all conflicts between 31.9% and 84.8%. Its low bias is not merely a denominator artefact—it makes the most texture decisions outright (31, against 22, 5 and 6). CLIP’s 96% rests on five texture decisions, so one flipped decision moves it 0.8 points. ResNet is therefore both the most colour-misled and the least shape-biased model, CLIP the most shape-dominant and least colour-sensitive. The two CLIP rows differ by two or three decisions, so cue preference is a property of the representation, not the readout.

Table 3: Mean cosine similarity between each clean feature and its transformed counterpart, paired per image. Comparable within a backbone, not across.

Backbone	cue conflict	grayscale	hue rotated	patch shuffled	translation 16px
clip_vit_b_32	0.680	0.893	0.894	0.738	0.968
resnet50	0.276	0.788	0.823	0.718	0.952
vit_b_16	0.258	0.739	0.689	0.652	0.974

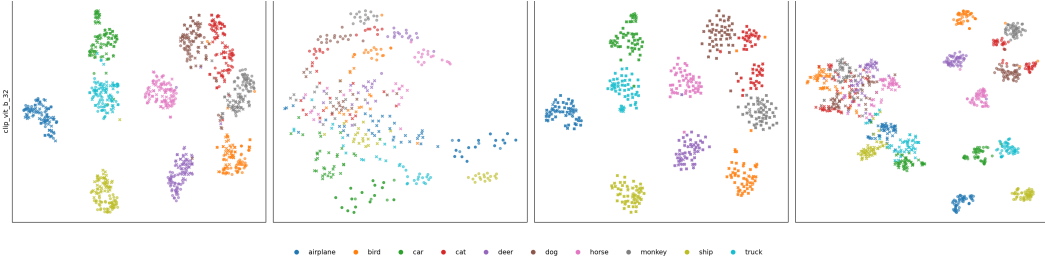


Figure 1: CLIP ViT-B/32 features, clean (circles) against transformed (crosses), coloured by ground-truth class; one t-SNE per panel on the combined conditions, so coordinates are not comparable across panels. The patch-shuffled panel is the mismatch discussed in the text. ResNet-50 and ViT-B/16 panels are in the repository.

3.2 Translation and patch structure

The largest accuracy change under translation is 1.2 points—six images—with several entries at 8 and 16 px above the unshifted baseline and the four directional means spanning 96.95 to 97.17, so the experiment cannot resolve a positional effect. Consistency falls monotonically, ResNet leading at 32 px (98.65 against CLIP zero-shot’s 97.30%) by too little to carry a claim. ViT’s consistency is non-monotonic (98.35, 98.90, 97.75%), and 16 px is exactly one ViT-B/16 token, preserving token contents and changing only positional embeddings where 8 px cuts through every token—the signature absolute positional embeddings predict, on about eleven predictions.

A 4×4 shuffle of 56 px patches costs ResNet 7.8 and ViT 8.4 points but CLIP 18.6 and 18.0. **The break is not CNN against transformer—ResNet and ViT differ by three images—but CLIP against the two ImageNet-supervised models.** Token size does not explain it: a 56 px patch spans about 3.5 ViT-B/16 tokens and 1.75 CLIP tokens, so CLIP’s larger tokens should preserve more local content, yet it loses twice as much. ResNet’s confidence falls only $0.895 \rightarrow 0.869$ and ViT’s $0.962 \rightarrow 0.900$ while both lose eight points, so both keep reading local evidence that survives permutation; CLIP zero-shot falls $0.943 \rightarrow 0.775$, in step with its accuracy.

3.3 Representation stability

Cosine stability is comparable within a backbone, not across. Ranked within each, ViT and CLIP order the transformations identically under both measures; ResNet does not. Hue moves its features *less* than grayscale (0.823 against 0.788) while flipping more predictions (27 against 18) and costing $2.2\times$ the accuracy: the smaller movement lies along directions the classifier reads.

The second disagreement is larger. Under patch shuffling CLIP has the *highest* cosine stability (0.738, against 0.718 and 0.652) yet loses by far the most accuracy. Absolute values cannot be calibrated—no random-pair baseline was computed—but Figure 1 shows CLIP’s shuffled features leaving their clean clusters for a class-mixed region, animals with animals and vehicles with vehicles. The shift is small in angle and still crosses class boundaries. Cue conflict gives the lowest cosine for every backbone (0.28, 0.26, 0.68) but is confounded: AdaIN re-synthesises the whole image and no $\alpha = 0$ reconstruction was computed.

CLIP’s head and zero-shot classifier read one frozen embedding, so their cosine stability is identical by construction and any behavioural difference is the readout alone. Their accuracy changes differ by at most 0.8 points under every intervention, their patch collapse is equal and their shape bias differs by one point: **the embedding, not the readout, determines these decisions.**

Table 4: Task 2 main comparison. Separability is a two-way probe on source-validation against target features, chance 50%; source-only already reaches 99.86%.

Method	Photo F1	Art F1	Cartoon F1	Mean src F1	Sketch acc.	Sketch F1	Δ Sketch acc.	Separability (%)
Source-only	95.61	90.45	95.05	93.70	65.18	64.15	0.00	99.86
DAN	94.94	89.64	94.20	92.93	56.91	55.31	-8.27	84.34
DANN	94.25	87.21	94.94	92.13	31.81	31.52	-33.37	99.86
CDAN	95.17	88.10	93.60	92.29	14.58	24.09	-50.60	100.00

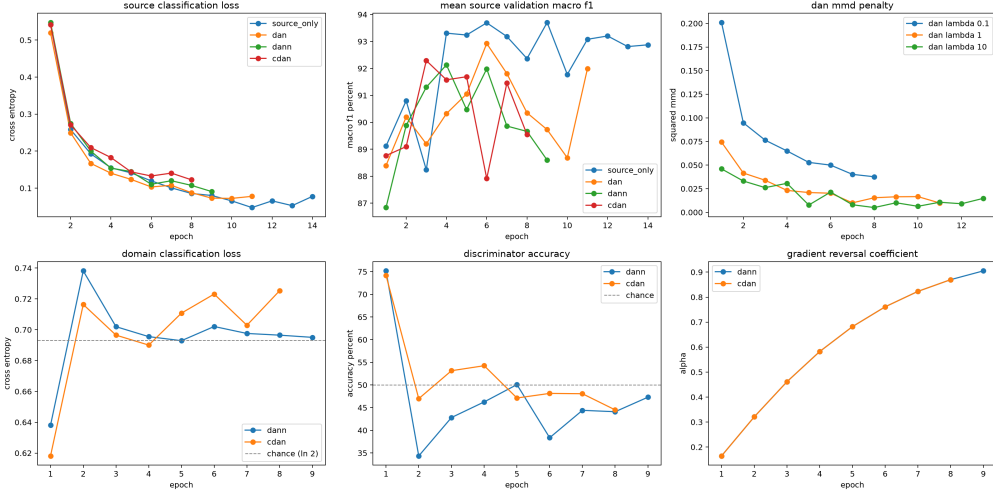


Figure 2: Task 2 training curves: source classification loss, mean source-validation macro-F1, the DAN MMD penalty at each λ_{MMD} , and the domain loss, discriminator accuracy and reversal coefficient for DANN and CDAN against their $\ln 2$ and 50% chance levels.

Attribution. ResNet-50 and ViT-B/16 share a corpus and objective, differing in architecture, capacity (25.6M against 86.6M) and augmentation recipe; that is the cleanest pair available, and their shape-bias difference is the only effect associated with architecture here, still confounded by capacity and recipe. ViT-B/16 against CLIP ViT-B/32 varies patch size, corpus, objective, embedding width and normalisation simultaneously, so CLIP’s patch collapse and 96% shape bias are not attributed to architecture—being a transformer is insufficient, since ViT-B/16 loses only 8.4 points on the same images. Two limits bear on the CLIP rows: the backbone was instantiated without QuickGELU, which is not the activation its released weights were trained with, so all CLIP numbers carry an unquantified offset; and per-image predictions were not retained, so the composition of undecided conflicts cannot be recovered. The weakest evidence here is the translation experiment, the second weakest any cross-backbone cosine comparison.

4 Task 2: Unsupervised Domain Adaptation

A ResNet-18 is fine-tuned on Photo, Art Painting and Cartoon and evaluated on Sketch, visible during adaptation as unlabelled images and labelled only after every checkpoint and setting is fixed. Four methods—source-only ERM, DAN, DANN and CDAN—share one training loop, initialisation and source batch sequence, so only the alignment term varies. Two probes of identical construction accompany the accuracy numbers: domain separability, and a class probe measuring how much class information survives in the same frozen features. **No method beats source-only ERM at its specified setting** (Table 4); the three failures divide into two mechanisms.

The unadapted gap. Source-only reaches 93.76% mean source-validation accuracy (Photo 96.11, Art 90.49, Cartoon 94.67) and 65.18% on Sketch: **a gap of 28.6 points**. Art is the weakest source for every method. The failures do not fall on the rare classes—house (80 images) and person (160) score 76.3 and 84.4%, while dog (23.8%, 184/772) and giraffe (37.7%, 284/753) are the worst and among the largest. The dominant confusions are all quadruped-to-quadruped (giraffe→horse 38.1%,

dog→horse 37.0%, dog→elephant 19.8%), and horse absorbs 35% of predictions against a true share of 20.8%, so its 88.7% recall is partly a consequence of acting as a sink. Once shading and colour are gone, four-legged silhouettes remain; guitar, with a distinctive outline, holds at 94.7%.

For two methods no alignment occurred. Across seven runs the rank correlation between separability and Sketch accuracy is $\rho = -0.58$ ($p = 0.18$). Source-only already sits at 99.86% separability—one image in 728 held-out rows—so alignment would have to move that far toward 50%. **Only DAN moved it** (84.34%, and 85.03 at $\lambda = 10$); DANN’s is 99.86%, identical to source-only, and CDAN’s 100%. Domain invariance does not account for their 33- and 51-point losses.

The discriminator and the probe measure orthogonal quantities. DANN’s domain loss sits at $\ln 2$ from epoch 3, its discriminator averages 43.5% over epochs 2–9 and its maximum domain logit never exceeds 0.48, so at the level the discriminator sees the domains are at chance. Deviation 3 reconciles this with near-perfect separability: the discriminator reads `F.normalize(features)` and so constrains direction only, while the probe is fitted on raw 512-d features whose mean norm grows from 44 to 108 during training. **The probe exploits exactly the channel the normalised domain loss cannot touch.**

Class information survived; the readout did not. A probe of identical construction predicting the seven classes from the same frozen features recovers 83.5–88.7% on Sketch in *every* run, against 88.6% for source-only, with source-side probes at 92–94.5%. The gap between what the features support and what the model achieves is 31.3 points for DAN, 51.6 for DANN and 70.0 for CDAN. The four-to-five-point degradation for DANN, CDAN and $\lambda = 10$ is real but far from destroyed class structure, so **“alignment removed class-discriminative information” is not supportable for any method here**: the adversarial objective damaged the readout on Sketch without making the domains less separable. The probe uses Sketch labels and is an oracle upper bound on linearly decodable structure.

Two failure modes. DAN is the only method with real alignment and its damage follows the label-shift pattern below. DANN and CDAN are optimisation failures: separability did not fall, source validation held (92.1 and 92.5%), the probes are intact, and predictions concentrate on one class—DANN sends 57.7% of Sketch predictions to house and CDAN 89.3% to person, with 96.8–99.5% of horse, dog and elephant going to person. Negative transfer appears at both levels: DAN loses 842 images source-only had right and gains 517, DANN loses 1,491 and gains 180, CDAN loses 2,022 and gains 34. Elephant, horse and guitar are damaged by all three; no class is improved by all three.

4.1 Class-conditional alignment

CDAN is below both marginal methods on every target metric (14.58% accuracy, 24.09 macro-F1), yet its class probe is within a point of DANN’s, so the difference lies in the readout. CDAN feeds the discriminator $\text{vec}(f \otimes p)$, placing the feature in the block of the predicted class. Once the classifier collapses onto person, $p \approx e_{\text{person}}$ for almost every target image and the feature occupies one block, so the discriminator separates domains by *which block is non-zero*—consistent with separability of 100% and domain logits of 0.6–1.3 against DANN’s 0.12–0.48. Because neither factor is detached, the reversed gradient also reaches the classifier head through p . Per-epoch target entropy was not logged, so this follows from the code and the recorded distributions rather than from measurement.

The result does not test whether conditioning preserves semantic structure: conditioning on a collapsed prediction vector is uninformative by construction, and the collapse was already present at the selected checkpoint, epoch 3. CDAN is also the only run whose macro-F1 exceeds its accuracy, by 9.5 points, because person reaches 100% recall on its 160 images and house 63.8% on its 80 while the five large classes are near zero.

4.2 Alignment strength

The expectation recorded before the sweep was read was that separability would fall steadily, source performance would hold then slip at $\lambda = 10$, and Sketch accuracy would peak in the middle. Source behaviour held (−5.1 macro-F1 at 10); the other two failed. Separability drops once then stays

Table 5: DAN at $\lambda_{\text{MMD}} \in \{0.1, 1, 10\}$. Source-validation selection chooses $\lambda = 1$ by 0.019 macro-F1, the worst of the three on target.

λ_{MMD}	Mean src acc.	Mean src F1	Separability	Target acc.	Target F1	Δ target acc.	Src-val pick
0.10	92.57	92.91	98.76	75.49	71.75	10.31	False
1.00	93.21	92.93	84.34	56.91	55.31	-8.27	True
10.00	87.93	87.82	85.03	68.44	67.28	3.26	False

flat, with $\lambda = 10$ five probe images above $\lambda = 1$, and Sketch accuracy is a V with its minimum at the specified setting (75.49, 56.91, 68.44). **The best target result in the study came from the weakest alignment**, at 98.76% separability—an essentially unaligned model. No mechanism for the non-monotonicity is established; seed variance, the different selected epochs (3, 6, 8) and the label-shift pull at $\lambda = 1$ remain candidates. At $\lambda = 10$ the penalty crowds out classification: loss 1.05 at epoch 1 then 0.32–0.58 against 0.05–0.10 elsewhere, gradient norms averaging 32–59 with a maximum of 425, clipping binding on every step, mean feature norm reaching 468, and the Sketch class probe down 4.4.

Source-validation selection picks $\lambda = 1$ by 0.019 macro-F1—about seventy times smaller than this run’s epoch-to-epoch jitter and twenty-five times smaller than the run-to-run noise of Section 8—selecting the worst of the three at a cost of 18.6 points. It can reject $\lambda = 10$, which loses five points on the source side, but cannot distinguish 0.1 from 1, which is where the target difference lies.

A bound the data imposes. Under the sampling used, person is 21.7% of the source mixture and house 14.5%, against 4.1% and 2.0% in Sketch; the total variation between the priors is 0.3007. Perfect marginal alignment equalises the predicted-label distributions, and the triangle inequality gives $\varepsilon_T \geq \text{TV} - \varepsilon_S$ [Zhao et al., 2019]: source error may be traded for target error, but the sum is pinned. With source-only’s 6.2% source error this caps Sketch accuracy at 76.2%, so DAN at $\lambda = 0.1$ reaching 75.49 against its own 77.4% ceiling is consistent; the 69.9% figure is the $\varepsilon_S = 0$ special case and applies to no model here. The bound reads target labels and was computed after the lock.

It explains part of the outcome. DAN’s predicted distribution moved toward the source prior (total variation 11.7 against 37.6), it over-predicts person and house, and its losses land where the mechanism says: horse −290 images, elephant −253, guitar −85, all under-represented in the source prior. **Its two largest gains, dog +198 and giraffe +94, are also under-represented**, so the mechanism does not cover them; those come from reduced absorption into horse, whose share falls from 34.8% to 13.0%. It does not apply to DANN or CDAN, which overshoot the prior rather than converging to it and reduced no separability.

The domain-weight arm. Reducing DANN’s domain-loss weight from 1 to 0.1, trained with the instructor’s approval as a stability diagnostic, moves Sketch accuracy from 31.81 to 67.80%—36 points, and 2.6 above source-only. Neither weight aligned the domains (99.86 and 99.73%), so the gain is less adversarial interference with the readout rather than better alignment. This concerns this configuration, not the method.

5 Task 3: Domain Generalization

Sketch is withheld entirely: no Sketch image is loaded by training, source-side diagnostics, checkpoint selection or hyperparameter choice. ERM is Task 2’s source-only checkpoint loaded unchanged, so the three methods differ only in objective. DAN-DG applies the same MMD implementation to the three unordered pairs of source domains; SAM removes no domain information and instead seeks a locally stable solution. Two source-side diagnostics run before the lock: a three-way source-domain separability probe (chance 33.3%) and the loss increase after one normalised gradient-ascent step of radius 0.05 on a fixed validation batch.

Both target-free methods beat the shared baseline. DAN-DG reaches 71.93% on Sketch (+6.74) and SAM 74.06% (+8.88), against sampling standard errors of about 0.7 points. **This is the reverse of Task 2, where every method had strictly more information and every method lost.**

Table 6: Task 3 main comparison. Separability is a three-way source probe, chance 33.3%, not comparable with Task 2. Sharpness uses the same radius SAM trains against.

Method	Mean src acc.	Mean src F1	Worst src acc.	Sketch acc.	Sketch F1	Δ Sketch acc.	Separability (%)	Sharpness
ERM	93.755	93.704	90.488	65.182	64.148	0.000	84.718	0.459
DAN-DG	93.550	93.314	90.000	71.927	71.920	6.745	68.439	0.428
SAM	96.193	96.171	93.659	74.065	75.327	8.883	85.714	0.130

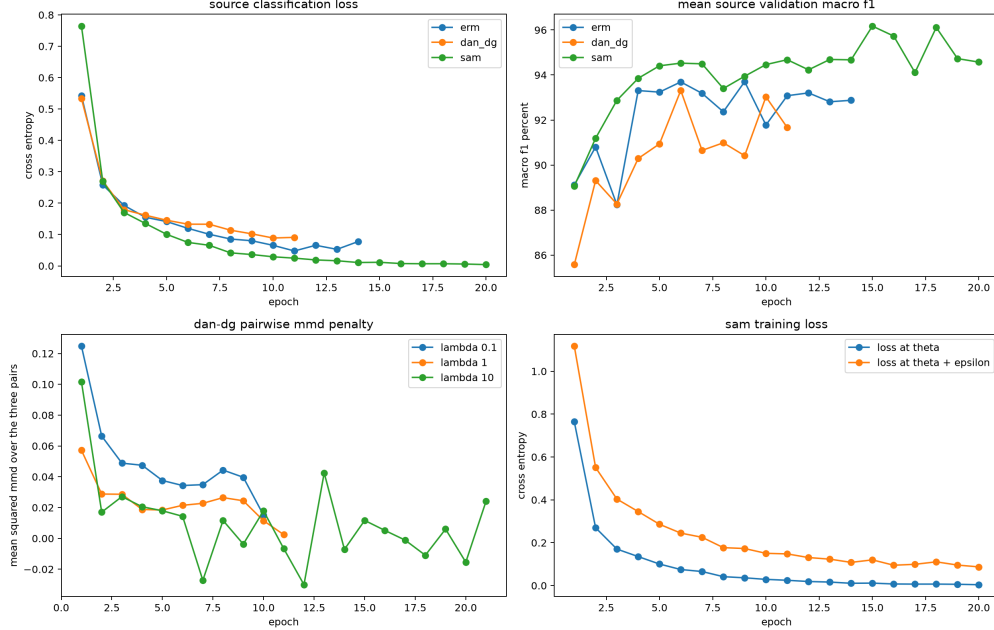


Figure 3: Task 3 training curves: source classification loss and mean source-validation macro-F1, the MMD penalty at each λ_{DG} , and SAM’s loss before and after its ascent step.

Source metrics rank the best model and cannot resolve the other two. SAM leads on mean source (96.19), worst source (93.66, Art in every case) and Sketch. ERM is ahead of DAN-DG by 0.2 points on the mean and 0.5 on the worst, both inside the noise band, yet DAN-DG leads by 6.7 on Sketch; source metrics therefore fail to resolve a real target difference rather than anti-correlating with it, and worst-source is the mean ranking shifted by about three points. Class-level failures are invisible from the source side in both directions: ERM’s Sketch dog accuracy is 23.8% against a source macro-F1 of 93.7, and DAN-DG improves on average while person collapses by 43.75 points—70 of 160 images, 48% of them predicted giraffe—with its source macro-F1 down only 0.39.

DAN-DG is the one case where invariance and target performance move together. Source-domain separability falls from 84.72% to 68.44%, covering 31.7% of the distance to the floor against a probe standard error of about 2.1, with the source side intact. Two things distinguish it from Task 2’s DAN: it aligns *labelled* domains, so every aligned feature also carries a class loss, and the label marginals of the aligned sets differ by a total variation of 0.11–0.18 rather than 0.30. The single casualty is consistent with the remaining mismatch—person is both the class whose prior varies most across the three sources (25.9, 21.9, 17.3%) and the most over-represented source class relative to Sketch. The data are consistent with this; it is not established.

5.1 Penalty strength and local stability

At $\lambda_{DG} = 10$ the model never learns: mean source accuracy is 32.9% and worst-source 22.4%, the classification loss sits between 1.72 and 1.97 for all 21 epochs against $\ln 7 = 1.95$, and at epochs 1–2 source accuracy matches what predicting one class for everything would score. The MMD penalty reaches zero by epoch 2 with 8 of 21 epoch means negative, as an unbiased estimator near zero should. **Its separability of 48.5%—the closest to chance anywhere here—is degeneracy, not invariance.**

Table 7: DAN-DG at $\lambda_{\text{DG}} \in \{0.1, 1, 10\}$. At $\lambda_{\text{DG}} = 10$ the model never learns, so its low separability is degeneracy rather than invariance.

λ_{DG}	Mean src acc.	Mean src F1	Worst src acc.	Separability	Sketch acc.	Sketch F1	Δ Sketch acc.
0.10	93.79	93.92	90.98	83.39	64.55	65.80	-0.64
1.00	93.55	93.31	90.00	68.44	71.93	71.92	6.74
10.00	32.94	28.30	22.44	48.50	35.30	27.59	-29.88

Table 8: Four post-hoc scores on the frozen vanilla model, all reading one cached set of logits and features. Rejection is measured at the validation-calibrated threshold.

Score	AUROC near	AUROC far	AUROC all	Known accept (%)	Near reject (%)	Far reject (%)
MSP	81.93	90.22	86.08	94.69	28.25	46.38
MLS	80.40	90.00	85.20	95.20	30.88	54.12
Energy	80.41	90.09	85.25	95.25	30.75	56.00
Mahalanobis	79.84	92.43	86.14	94.63	28.25	52.75

Whether alignment removed class information or prevented it from being learned cannot be settled: the loss was at $\ln 7$ from the first epoch and Task 3 has no class probe.

SAM reduces the sharpness proxy, but the diagnostic is near-circular. The ordering is SAM $0.130 < \text{DAN-DG } 0.428 < \text{ERM } 0.459$, matching the Sketch ordering. Adding the $\lambda_{\text{DG}} = 0.1$ run breaks it—second flattest at 0.223, worst working model on Sketch at 64.5%—taking the rank correlation over four working models to -0.40 . The proxy is $\mathcal{L}(\theta + 0.05 g/\|g\|) - \mathcal{L}(\theta)$, and SAM trains to minimise that quantity at that radius with the same normalised ascent, so its reduction shows SAM optimised its own objective and that this transfers to held-out source images. Base losses differ threefold (0.059–0.176); normalising by them keeps SAM flattest. At $\lambda_{\text{DG}} = 10$ the proxy reads 40.3, so the measurement is not meaningful away from a low-loss solution.

SAM’s advantage is not cleanly attributable to flatness. It also improved the source side (+2.4 mean, +3.2 worst), and its source-domain separability (85.71) is within probe error of ERM’s (84.72), so **the best domain-generalization model came with no change in domain invariance**. The competing explanation is better optimisation: SAM performs two forward/backward passes per step, trained 20 epochs against ERM’s 14, and at the matched epoch 9 its source advantage is +0.24. Separating the two would need a compute-matched ERM or multiple seeds.

Unlabelled target access. Against the shared ERM baseline, DAN with unlabelled Sketch reaches 56.91% and DAN-DG without it 71.93%: **the target-aware method is 15.0 points worse than the target-free one, and worse than doing nothing**. Target access nonetheless *can* help: Task 2’s DAN at $\lambda = 0.1$ produced the best Sketch result in the study, at a setting source-validation selection rejected by 0.019 macro-F1. The comparison is not causal, since target access co-varies with what is aligned, the MMD sample size (24 against 24, versus 8 against 8 per pair), the bandwidth pool, whether the aligned samples carry a class loss, and the label-marginal mismatch; optimiser, budget, initialisation, batch order, stopping length (both 11 epochs, best epoch 6) and the MMD code are identical. It therefore bears on this use of unlabelled Sketch, not on unlabelled target data in general.

6 Task 4: Open-Set Recognition

CIFAR-10 supplies ten known classes; sixteen fixed CIFAR-100 test classes supply eight near and eight far unknowns of 800 images each. Four post-hoc scores are computed on one frozen vanilla model from a single cached set of logits and features: maximum softmax probability, maximum logit, energy and Mahalanobis distance. GCSC is the vanilla recipe with RandAugment inserted and nothing else changed; PROSER fine-tunes from the vanilla checkpoint with five dummy classifiers and manifold-mixup data placeholders [Zhou et al., 2021]. Every threshold is the 95th percentile of unknownness on CIFAR-10 validation data.

Semantic distance governs detectability, and the best score depends on which kind. Every score separates far unknowns better than near ones, by 8.3 to 12.6 AUROC points, and the ranking

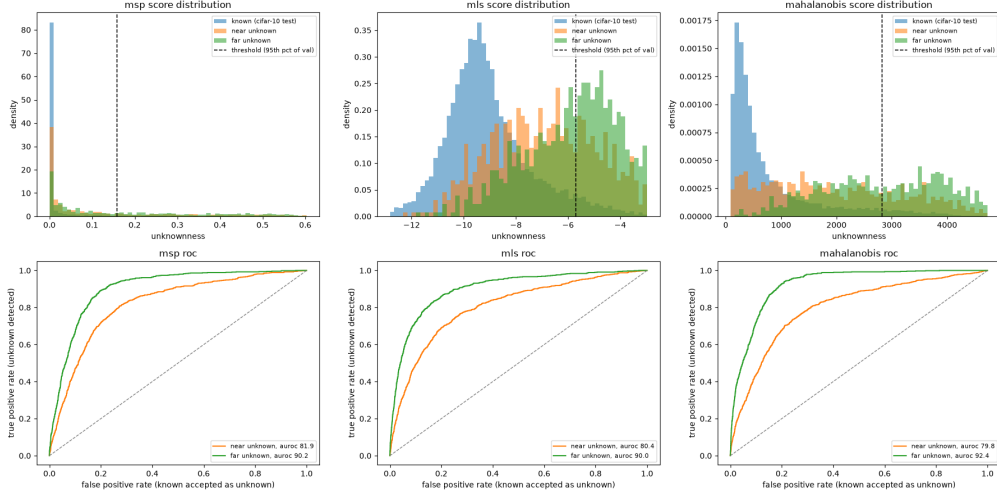


Figure 4: Score distributions for knowns against near and far unknowns, with the corresponding ROC curves, for MSP, MLS and Mahalanobis on the frozen vanilla model.

Table 9: Accepted unknowns under the vanilla MLS threshold. A score further below the threshold is a more confident acceptance.

Group	Unknown class	Predicted known	Score	Threshold
near	wolf	dog	-7.567	-5.711
near	tractor	truck	-7.792	-5.711
near	pickup_truck	automobile	-6.777	-5.711
far	mushroom	cat	-11.068	-5.711
far	bowl	frog	-9.693	-5.711
far	wardrobe	truck	-5.839	-5.711

reverses between them: MSP is best on near (81.9) and worst-equal on far, Mahalanobis worst on near (79.8) and best on far (92.4). Only those two leads exceed roughly one standard error. Feature distance sees semantically distant inputs because they occupy different regions of feature space, while near unknowns sit *inside* known-class regions where distance says little. MLS and Energy differ by 0.02 and 0.08 AUROC points—at 99.98% training accuracy the softmax is sharply peaked, so log-sum-exp reduces to the maximum logit.

The operating point changes the picture. At 95% known acceptance, vanilla MLS accepts 69% of near unknowns—553 of 800—as known classes, and 45.9% of far unknowns. MSP wins near AUROC yet is worst-equal on near rejection and 7.8 points worse on far, its validation distribution being saturated near zero unknownness and leaving no resolution at the 95th percentile. Achieved acceptance varies between 94.63 and 95.25% across scores, so the rates are compared at marginally different operating points.

Near unknowns go to truck (155), cat (104) and automobile (99)—65% in the top three—and far unknowns to cat (80), bird (74) and truck (66). The rankings are dissimilar ($\rho = 0.125$), but cat and truck are heavy in both. The near group’s absorptions split 274 to vehicles against 279 to animals, matching its composition; horse absorbs only 18 despite camel being a plausible neighbour.

Wolf predicted dog, tractor predicted truck and pickup truck predicted automobile are semantically unremarkable, each 1.1 to 2.1 below threshold. Wardrobe predicted truck is visually plausible and marginal at 0.13; bowl predicted frog is plausible on colour and texture. **Mushroom predicted cat has no semantic account and is the most confident of the six**, 5.36 below threshold—more confident than a typical known test image, so no logit-confidence threshold near this operating point could reject it without rejecting many knowns.

Table 10: Vanilla, GCSC and PROSER under MLS, with PROSER’s placeholder score as an additional row. Closed-set accuracy uses the ten known logits throughout.

Model	Score	CSA (%)	AUROC near	AUROC far	AUROC all	Near reject (%)	Far reject (%)
Vanilla	MLS	94.88	80.40	90.00	85.20	30.88	54.12
GCSC	MLS	95.01	80.62	91.39	86.01	32.75	59.38
PROSER	MLS	94.30	79.83	88.67	84.25	28.00	51.38
PROSER	Placeholder	94.30	77.87	89.07	83.47	25.62	51.00

6.1 Augmentation and placeholders

Augmentation improved rejection without improving classification. GCSC’s closed-set accuracy moves +0.13 points, below the binomial standard error of 0.22. Its AUROC gains are +0.23 near and +1.39 far, and far rejection gains 5.25 points—about three standard errors—at unchanged known acceptance. **The far gain is roughly six times the near gain in AUROC.** RandAugment kept training loss away from zero (0.032 against 0.0024 at epoch 100; training accuracy 98.98 against 99.98%), which plausibly yields less over-confident logits far from the training manifold while leaving near unknowns inside known-class regions; this is not measured. Closed-set and open-set improvements therefore did not move together.

PROSER was worse on every measure, and the selection rule explains why. It costs 0.58 points of closed-set accuracy and loses 0.57 and 1.33 AUROC points near and far, and its placeholder score is 1.96 points worse than plain MLS on the same model on near unknowns, where it should help most. **The selected checkpoint is epoch 4 of 50**, where the data-placeholder cross-entropy is 1.975 against a chance value of $\ln 11 = 2.40$. The classifier-placeholder term converged by epoch 2, but the data-placeholder term only begins training around epochs 6–7, exactly when validation accuracy starts falling; selecting on CIFAR-10 validation accuracy therefore picks the checkpoint least changed by the data-placeholder objective. This follows from the specified rule rather than from the implementation, but its cost is that the mechanism PROSER exists for was barely present in the evaluated model. PROSER also differs from vanilla in 50 extra epochs, a different learning rate and BatchNorm statistics updated on mixed features, with no “vanilla plus 50 epochs” control.

Where interpolation is insufficient. Data placeholders populate the gaps *between* known classes, but the failures sit *inside* single class regions with large logits: mushroom predicted cat at a maximum logit of 11.07 and bowl predicted frog at 9.69 lie between no two CIFAR-10 classes, and wolf predicted dog sits inside the dog region. Interpolation between knowns cannot produce a proxy for an unknown arriving from a direction no pair of known classes spans.

7 Cross-Task Discussion

Visual cues and distribution shift. Task 1’s backbones lose about two points to grayscale and at most 1.2 to translation, but up to eighteen to global scrambling; the PACS shift removes colour, texture and fill while preserving contour, and costs 28.6 points. Task 1’s colour results predict a small gap and the observed gap is large, because no Task 1 intervention reproduced the sketch shift—grayscale keeps shading and local texture, which a line drawing discards, while global arrangement, Task 1’s strongest fragility, is what sketches preserve. The shared vulnerability is narrower: once surface evidence is gone, coarse silhouette separates guitar, house and person but not four-legged animals, and every Task 2 and Task 3 model’s dominant confusions are quadruped-to-quadruped. The two settings differ in dataset, backbone and training regime, so the comparison is qualitative.

Invariance versus discriminability. Across every (separability, target accuracy) pair in both tasks—read separately, the probes having different chance levels—there is no consistent relationship. Exactly one run shows invariance increasing while target performance improves and the source side holds, DAN-DG at $\lambda_{DG} = 1$, and it still cost one class 43.75 points. Two runs show target performance collapsing with separability unmoved, DANN and CDAN, where the class probe shows the features intact and “invariance removed discriminability” is the wrong diagnosis because no invariance occurred. The clearest illustration of the failure the specification warns about is $\lambda_{DG} = 10$, where the evidence says the model never learned rather than that alignment removed something. **Across seven**

alignment runs invariance was neither necessary nor sufficient: the best target model in each task came with essentially no measurable reduction in separability, and the most invariant models were among the worst.

Recognition versus rejection. Closed-set accuracy and open-set AUROC move in the same direction across the three Task 4 models, but every closed-set delta is at most 0.6 points against a standard error of 0.22. GCSC gains five points of far rejection at flat accuracy, so the improvement did not come from a better classifier. The same separation appears in Task 2, where class structure survived in representations whose classifiers had collapsed, and here, where a 94.9%-accurate representation accepts 69% of near unknowns. Among the configurations tested, GCSC with an MLS or Energy threshold is the one to deploy where a confident error is costly, and even then two thirds of near unknowns are accepted.

Synthesis. A representation should preserve the class-discriminative structure that survives surface change: shape held up across Task 1’s colour and position interventions, and Task 2’s probe recovered 83–89% of Sketch classes from features whose classifiers produced 15–32%. The variation it should suppress—rendering style, colour, texture—is clear, but *forcing* that invariance through a discrepancy or adversarial loss did not reliably produce it: adversarial forcing never achieved it, strong marginal forcing prevented class learning, and the two methods that improved on an unseen domain either aligned already-labelled domains or changed no invariance at all. Rejection of inputs outside the label space is unreliable for every method tested, worst for the near inputs most likely to arrive, and occasionally failing with above-typical confidence. Of the seven specified methods evaluated, two clearly beat their baseline, one marginally, and four lost.

8 Limitations

Every experiment ran once, so **no result carries an error bar**, and without `cuda.deterministic` runs are not bitwise reproducible: Task 2’s source-only run, executing byte-identical code, trained 23 epochs on one occasion and 14 on another, with mean source-validation macro-F1 differing by roughly half a point. **Differences below about one point should not be read as effects**, which removes Task 1’s translation result, GCSC’s closed-set accuracy, and most score-to-score near-unknown differences.

Three deviations changed what the methods optimise, so every DAN, DANN, CDAN and DAN-DG conclusion concerns the modified objective: gradient clipping, which binds on 77–100% of steps and is therefore part of the optimiser in every run including ERM; the unbiased MMD estimator; and the normalised discriminator input, whose interaction with a probe fitted on raw features is analysed in Section 4. The class probe is trained on target labels and is an oracle upper bound on linearly decodable structure. Task 1’s CLIP backbone was instantiated without QuickGELU, so all CLIP numbers carry an unquantified offset, and per-image predictions were not retained. Target labels were read once on the selected checkpoint, so the gap between it and the best-transferring epoch is not measurable. Several proposed mechanisms—the non-monotonic λ sweep, CDAN’s degenerate conditioning, the absorption of unknowns by cat and truck—are consistent with the recorded diagnostics but were not measured directly.

9 Conclusion

The methods that improved on their own baselines were those that improved or regularised the supervised solution—alignment between labelled source domains, sharpness-aware optimisation, stronger positive augmentation—while those imposing a representational property by adversarial or strong marginal alignment did not, and in two cases damaged the classifier without achieving that property. The diagnostics are what make this statement measurable rather than speculative: a separability probe showing alignment never happened, a class probe locating the failure in the readout, and a label-shift bound showing what the dataset makes impossible regardless of method. A metric at one level licenses no claim at another: a discriminator at chance is not an aligned representation, low domain separability is not preserved class structure, and a 94.9%-accurate classifier is not a safe one.

References

- Robert Geirhos, Patricia Rubisch, Claudio Michaelis, Matthias Bethge, Felix A. Wichmann, and Wieland Brendel. ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness. In *International Conference on Learning Representations (ICLR)*, 2019.
- Xun Huang and Serge Belongie. Arbitrary style transfer in real-time with adaptive instance normalization. In *IEEE International Conference on Computer Vision (ICCV)*, 2017.
- Han Zhao, Rémi Tachet des Combes, Kun Zhang, and Geoffrey J. Gordon. On learning invariant representations for domain adaptation. In *International Conference on Machine Learning (ICML)*, 2019.
- Da-Wei Zhou, Han-Jia Ye, and De-Chuan Zhan. Learning placeholders for open-set recognition. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.